For my machine learning project, I will develop a “Movie Success Classifier” that predicts whether a movie will be successful or not successful based on key attributes such as: budget, runtime, genre, and popularity. The application will be built as a Graphical User Interface (GUI) using Python and the scikit-learn library. The goal of this project is to demonstrate how machine learning can identify patterns in movie data and use them to make predictions about future outcomes in the film industry.

The dataset for this project will be sourced from Kaggle, specifically the “TMDB 5000 Movie Dataset” (https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata) This dataset includes thousands of movies and contains useful features such as budget, revenue, genres, runtime, and popularity scores. To convert this into a classification problem, a new target variable will be created where a movie is labeled as “successful” if its revenue is greater than its budget, and “not successful” if its revenue is less than or equal to its budget. (If I have time, I would plan to implement that in addition, the profit exceeding a certain margin (ex: 10%, 20%, 30%, etc. of initial budget)). This transformation allows the dataset to be used for binary classification.

The machine learning model used in this project will be a Decision Tree Classifier. This model is well-suited for structured data and provides clear decision-making logic based on feature values. The process will begin by loading and cleaning the dataset using pandas, including handling missing values and removing irrelevant data. Categorical variables such as genre will be converted into numerical values so they can be used by the model via encoding. After preprocessing, the dataset will be split into training and testing sets, and the Decision Tree model will be trained on the training data. The model will then be evaluated using accuracy on the test data to determine how well it predicts movie success.

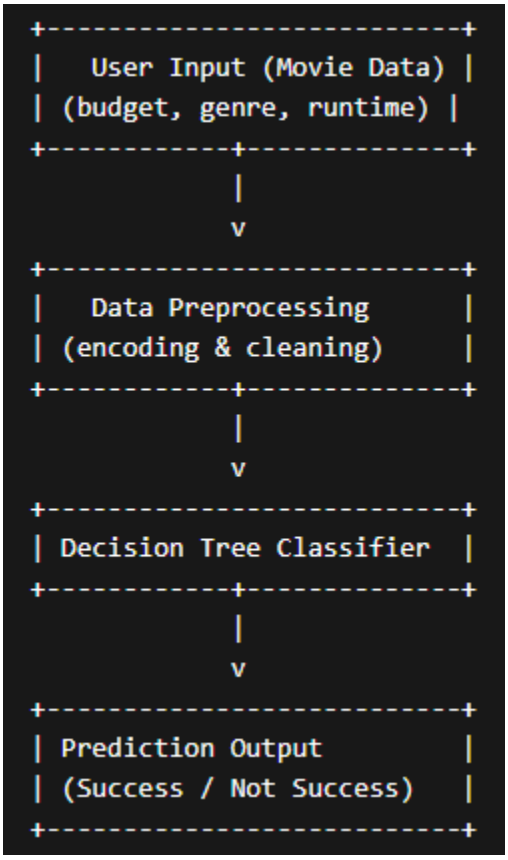
The system will also include a simple graphical user interface built with Tkinter. This interface will allow users to input movie details such as budget, runtime, and genre. After submitting the information, the trained model will process the input and display a prediction indicating whether the movie is likely to be successful or not. This makes the project interactive and user-friendly rather than only running in the background.

The overall system can be described as a flow where user input is first collected through the GUI, then passed through data preprocessing steps such as encoding and cleaning. After preprocessing, the data is sent to the trained Decision Tree model, which then generates a prediction. Finally, the result is displayed back to the user as either “Successful” or “Not Successful.”

This project meets all the required criteria because it uses a real dataset from Kaggle that has not been used in prior assignments such as Titanic or Iris, it applies a machine learning

classification model using scikit-learn, and it includes a graphical user interface for user interaction. Additionally, it involves meaningful data preprocessing and feature engineering, making it slightly more advanced than basic introductory projects.

Overall, this project demonstrates how machine learning can be applied in the entertainment industry to predict movie outcomes based on historical data. It combines data analysis, classification modeling, and user interface design into a single practical application.



(This image was made using ai to visualize the workflow)

For simplicity's sake, the layout of the GUI would be very similar to our titanic demo; although, I would not be opposed or outright changing the appearing as things developed